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EDUCATION

Exp. 2026 **Bachelor of Science in Mathematics**
University of Chittagong, Bangladesh

PROFESSIONAL EXPERIENCE

Dec 2025–Present **Research and Development Engineer**
NorthAxis, Remote
Lead applied research and technical discovery for core AI product features.

- Spearhead the research, prototyping, and evaluation of advanced AI models to drive new capabilities.
- Design and experiment with autonomous agentic workflows and retrieval-augmented generation (RAG) pipelines using LangChain.
- Train and fine-tune ML models and open-source LLMs to optimize for domain-specific tasks.
- Conduct feasibility analyses and benchmark state-of-the-art architectures for practical integration.

RESEARCH

UNDER REVIEW

2025 **CPU-Constrained Deep Learning for Tomato Disease Detection: Traditional, Modern, and Hybrid CNN Comparison**
With Lipon Chandra Das, Arnab Aich, Abu Saiman Md Taiham, and Atif Ibna Latif
Under Review, Springer Book Proceedings (24th International Mathematics Conference)

CONFERENCE PRESENTATIONS

2026 **Pathways from Digital Distractions and Study Habits to Academic Performance: Self-Regulation as Mediator Among Bangladeshi University Students**
With Md. Ali Arman Rafi · Scholars' Convergence Research & Innovation Summit (SCRIS), Chittagong, 2026

2026 **Quantifying the Congestion Premium: Disentangling the Determinants of Non-Metered Fares in a Developing Port City**
With Sifatul Islam · Scholars' Convergence Research & Innovation Summit (SCRIS), Chittagong, 2026

TECHNICAL SKILLS

Machine Learning & AI:

PyTorch, TensorFlow, Computer Vision (CNNs, ViTs), LangChain, Scikit-Learn

Backend & Data Architecture:

Python, FastAPI, Flask, C++, PostgreSQL, Firebase, NumPy, SciPy, Pandas

Frontend & Infrastructure:

TypeScript, JavaScript, Next.js, React.js, Git, L^AT_EX

HONOURS & AWARDS

2025 **Best Presenter — 24th International Mathematics Conference**
Awarded for the presentation of *CPU-Constrained Deep Learning for Tomato Disease Detection*.
References available upon request.